Triangle Number Sums

Time limit: 1.00 s
Memory limit: 512 MB

CSES 3406 | Mathematics | https://cses.fi/problemset/task/3406

A triangle number is a positive integer of the form $1+2+\ldots+k$. The first triangle numbers are 1, 3, 6, 10 and 15.

Every positive integer can be represented as a sum of triangle numbers. For example, 42=21+21 and 1337=1326+10+1.

Given a positive integer n, determine the smallest number of triangle numbers that sum to n.

Input

The first line has an integer t: the number of tests.

After that, each line has a positive integer n.

Output

For each test, print the smallest number of triangle numbers.

Constraints

- $1 \le t \le 100$
- $1 \le n \le 10^{12}$

Example

Input
5 1 2 3 42 1337

Output
1 2 1 2 3